

Guideline Title: Abdominal Aortic Aneurysm

Guideline Number: GUID-3203-M001

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Recognition of the patient with an aortic aneurysm that is expanding, ruptured, or caused an aortic dissection; provide and maintain adequate oxygenation, ventilation, and end-organ tissue perfusion; provide expeditious transport to the nearest appropriate medical facility.
- II. **DEFINITIONS:**
 - a. AAA - Abdominal aortic aneurysm
- III. **DEPARTMENT GUIDELINE:**
 - a. Establish care as defined by *GUID3203-G002 – General Patient Care*.
 - b. Establish two large-bore peripheral intravenous catheters in patients who are symptomatic.
 - c. Symptomatic, expanding AAAs may be managed conservatively unless directed by the accepting physician.
 - i. If the patient has pain and/or anxiety, treat as per *GUID3203-G005 - Analgesia and Sedation Management* or *GUID3203-G006 – Management for the Anxiety/Combative Patient*.
 - ii. If the patient remains tachycardic and hypertensive after analgesia and anxiolytics consider short-acting anti-hypertensives per *GUID3203-M010 - Hypertensive Emergency*.
 - d. Ruptured AAAs are a surgical emergency
 - i. If patient displays signs of inadequate perfusion in the setting of a ruptured aortic aneurysm, begin resuscitation with NS per *GUID3203-M011 - Hypotension*. Resuscitate to SBP of 80-100 mmHg.
 - ii. Allow permissive hypotension to maintain patient mental status. Limit fluid infusion. Over resuscitation may result in increased bleeding.
 - iii. Consider transfusion of blood products per *GUID3203-PR002 - Blood Product Administration*.
 1. Reversal agents should also be considered for patients who are anticoagulated.
 - e. If patient has a contained ruptured AAA or an aortic dissection, treat per *GUID3203-M018 Thoracic Aortic Dissection*
 - i. Goals: adequate end-organ perfusion with SBP < 120 mmHg and HR < 60 bpm.
 - ii. If SBP > 120 mmHg and HR > 60 bpm administer Beta Blockers

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1. **Esmolol bolus 500 mcg/kg/min IV** over 1 minute followed by **Esmolol infusion at 50 mcg/kg/min**. Titrate every 5 minutes by repeating 500 mcg/kg/min bolus over 1 minute and then increasing infusion by 50 mcg/kg/min. Maximum dose 300 mcg/kg/min
2. **Labetalol 10-20 mg IV** over 2 minutes. May repeat every 10 minutes until goal blood pressure obtained or maximum total dose of 300mg.
3. If SBP remains > 120 mmHg after β -blockers, or HR < 60 bpm prior to Beta Blockers administer:
 - a. **Nicardipine infusion at 5 mg/h**. Increase by 2.5 mg/h IV every 5-15 minutes to maximum of 15 mg/h.
 - b. **Nitroprusside (50 mg/250ml D5W) infusion at 0.3 mcg/kg/min**. The infusion can be increased by 0.5 mcg/kg/min every 5 minutes until desired BP is reached. Max dose 10 mcg/kg/min (Avoid in renal failure).
 - i. Treat reflex tachycardia from nitroprusside with labetalol or esmolol as above
- f. As the management of the patient with a symptomatic AAA is operative and extremely time dependent, the transport team should minimize scene time by performing all non-critical interventions while en route to receiving facility.

IV. DOCUMENTATION:

- a. EMS Charts

V. REFERENCES: